

Decision Trees

Decision Trees

A Decision Tree is a supervised machine learning algorithm used for both classification and regression tasks. The key idea is to split the dataset into smaller subsets based on feature values, and this process continues until a stopping condition is met (for example, no further splitting is possible, or maximum depth is reached). The result is a tree-like structure where :

Nodes represent tests on an attribute (feature).

Branches represent the outcome of those tests.

Leaves represent class labels (for classification tasks) or continuous values (for regression tasks).

Decision Node

Root Node

Sub-Tree

Decision Node

Decision Node

Leaf Node

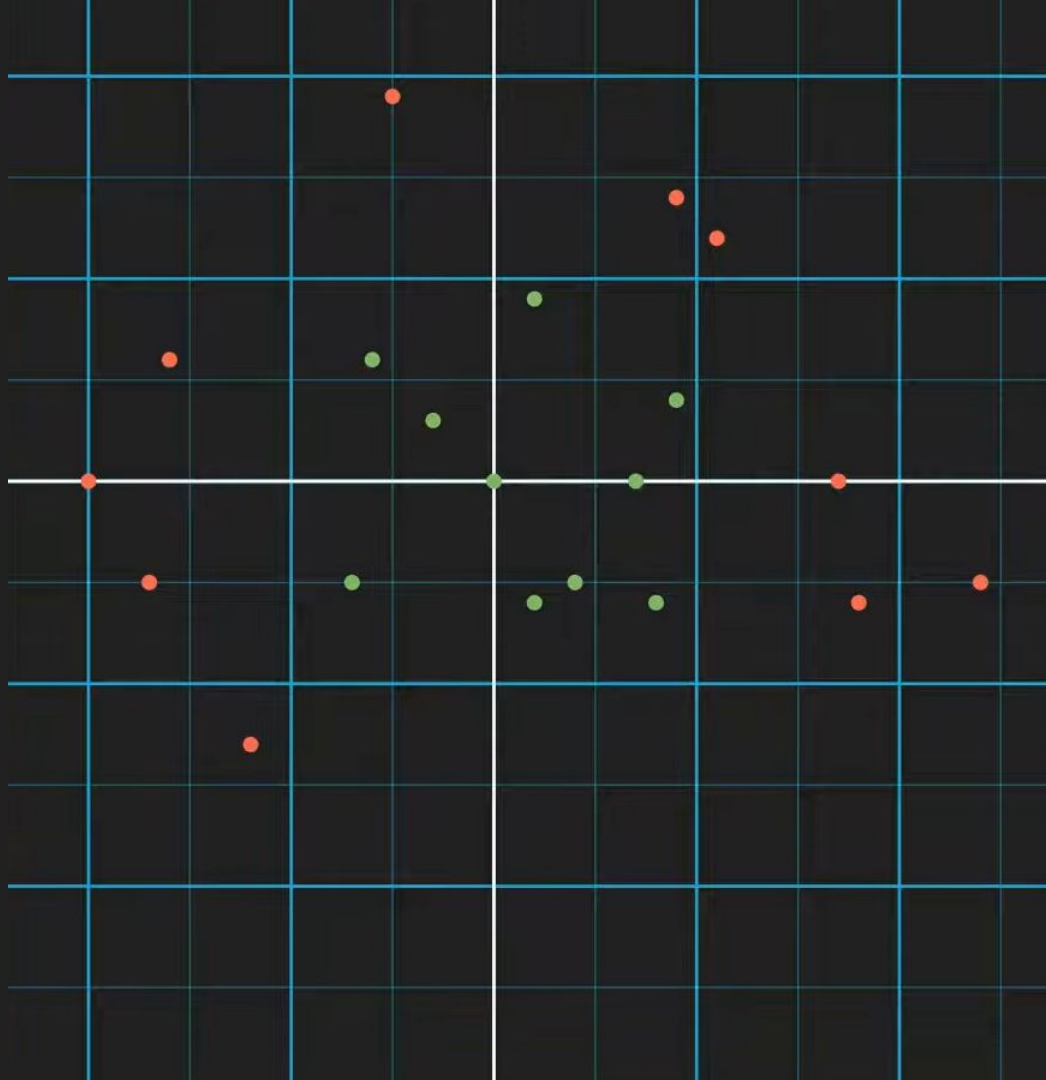
Leaf Node

Leaf Node

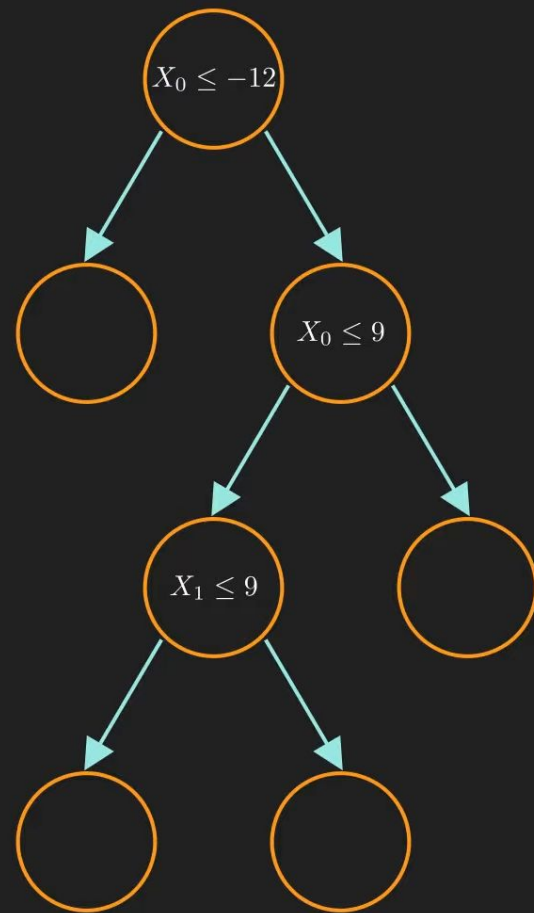
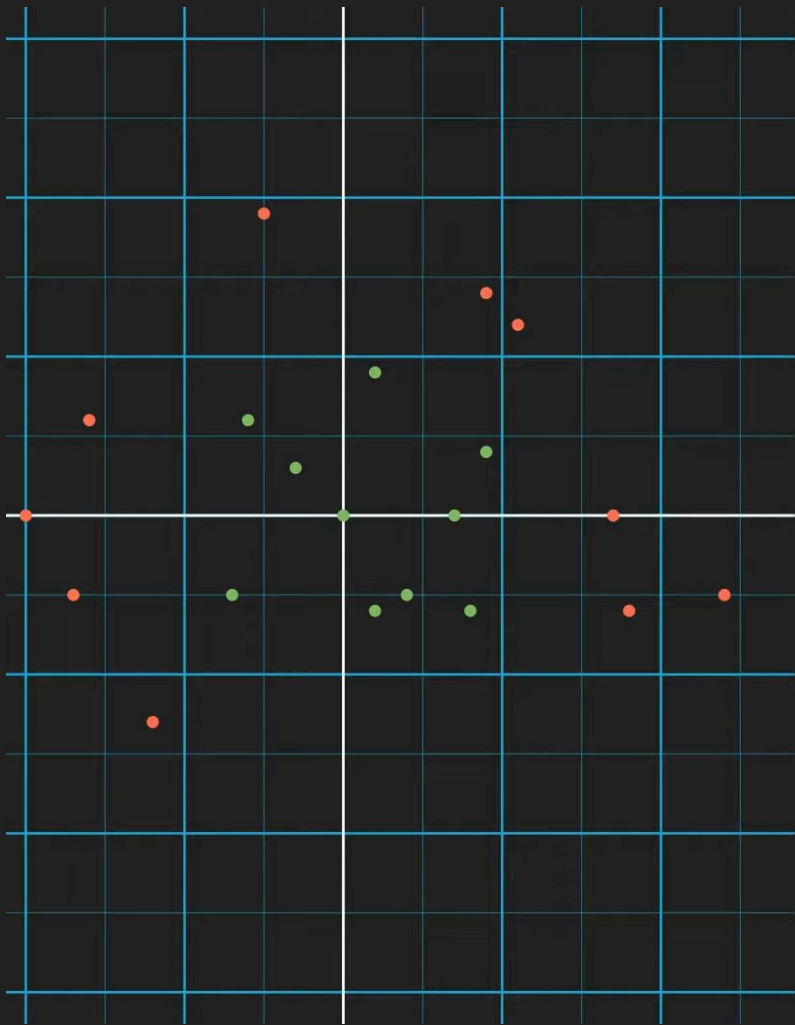
Decision Node

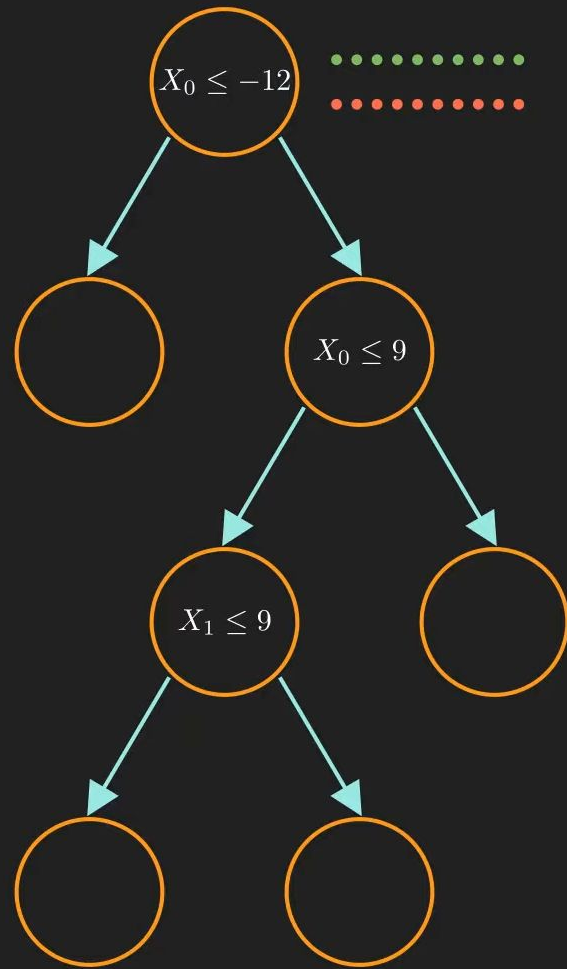
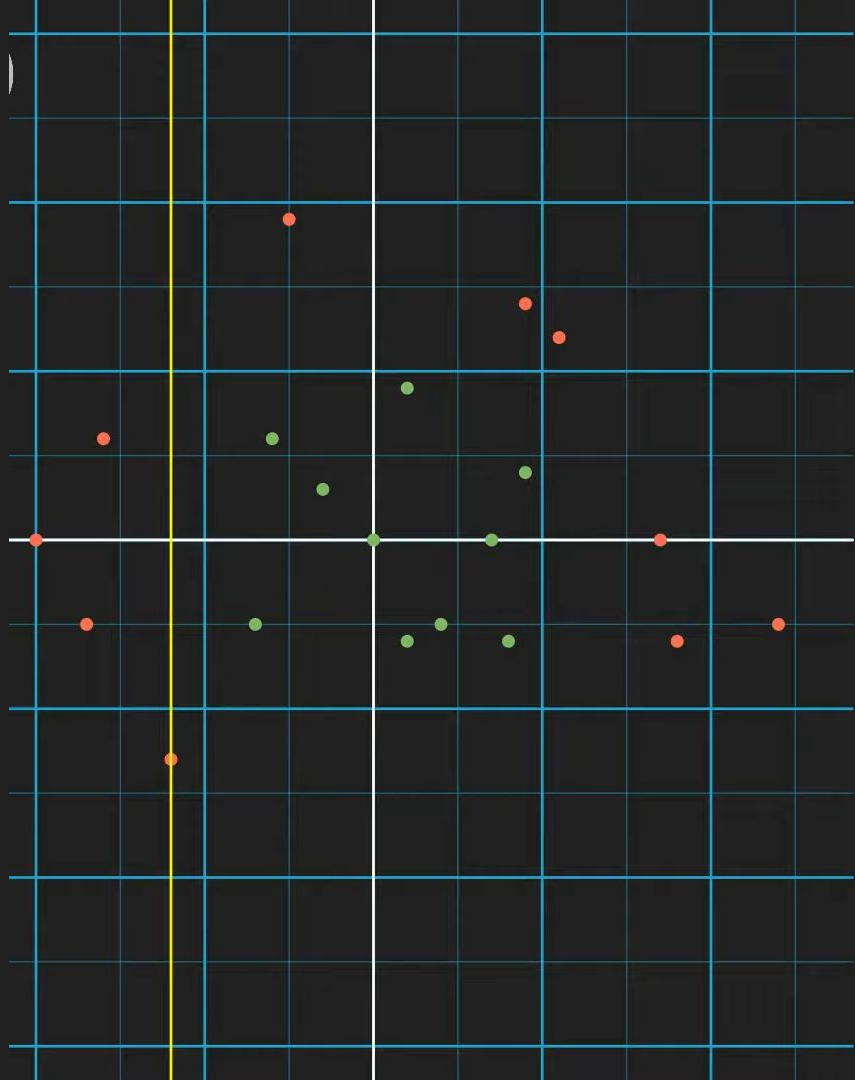
Leaf Node

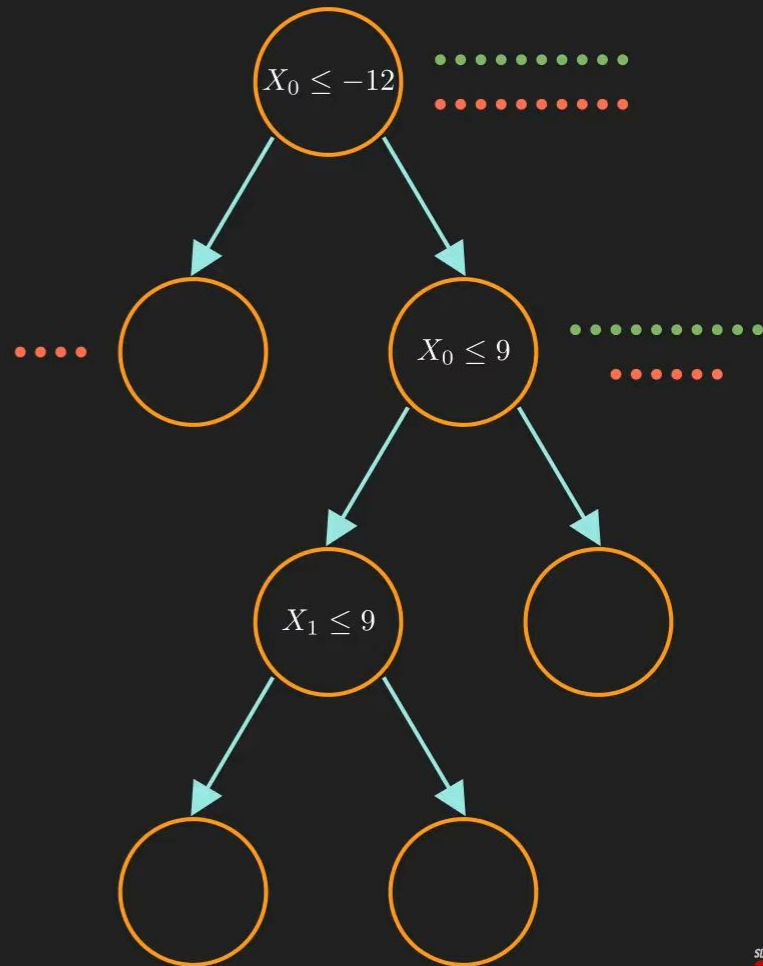
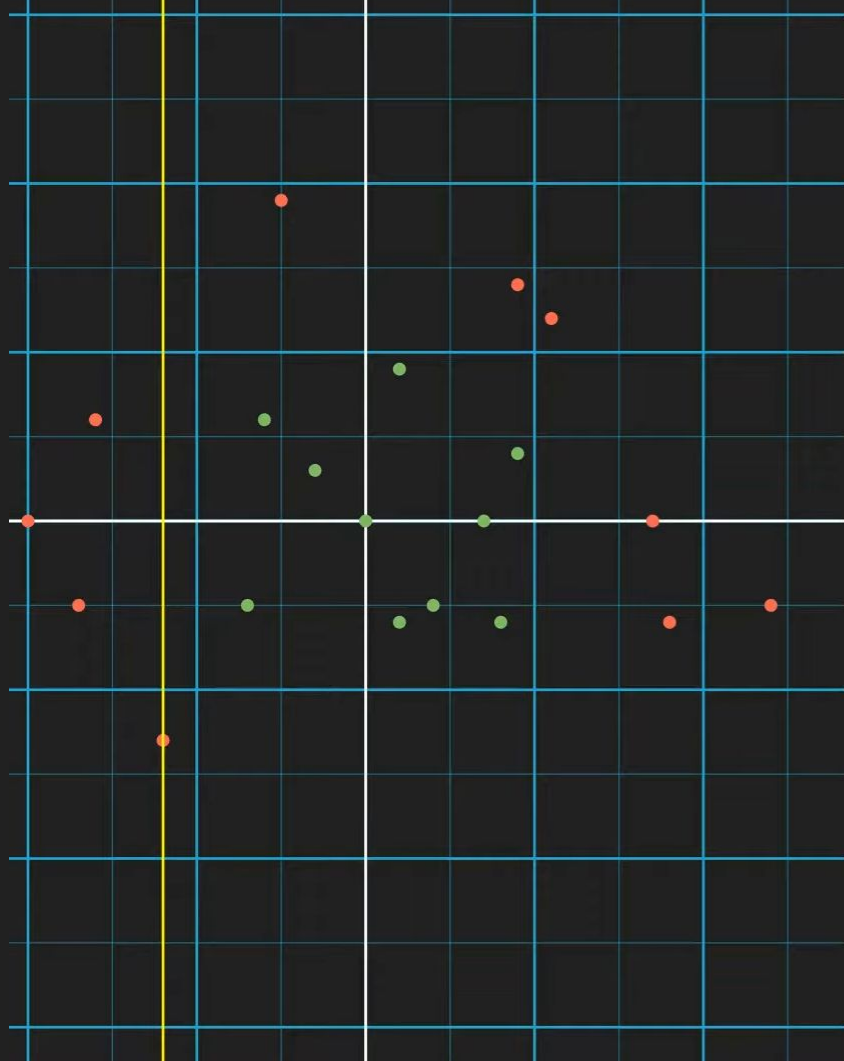
Leaf Node

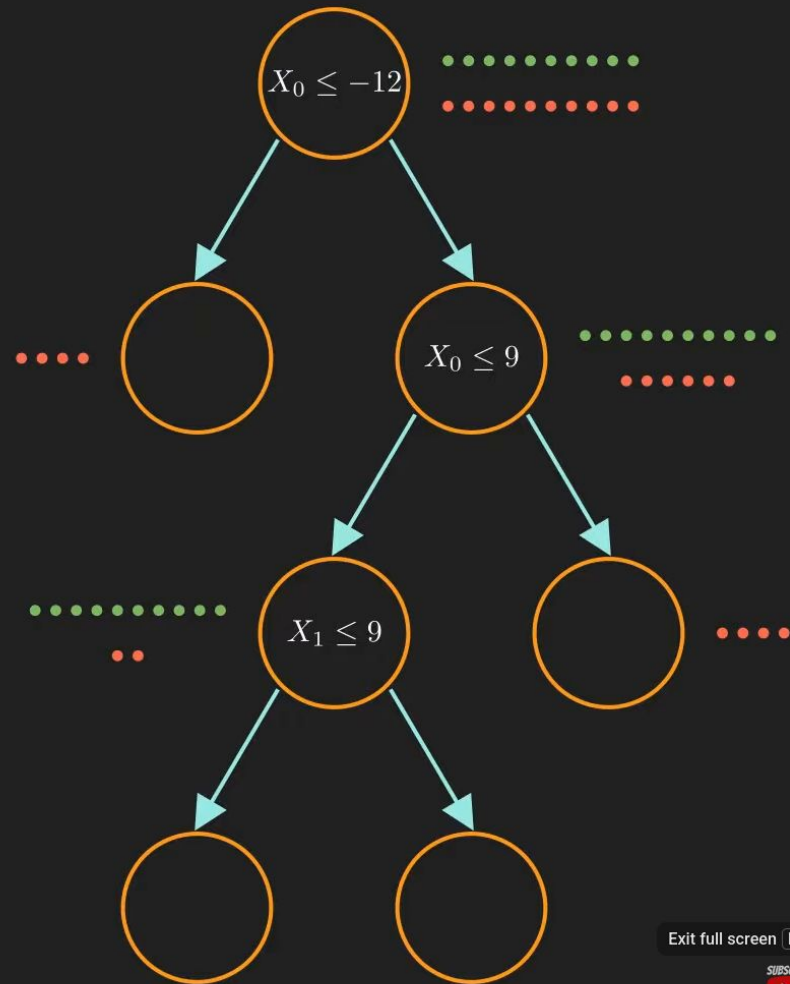
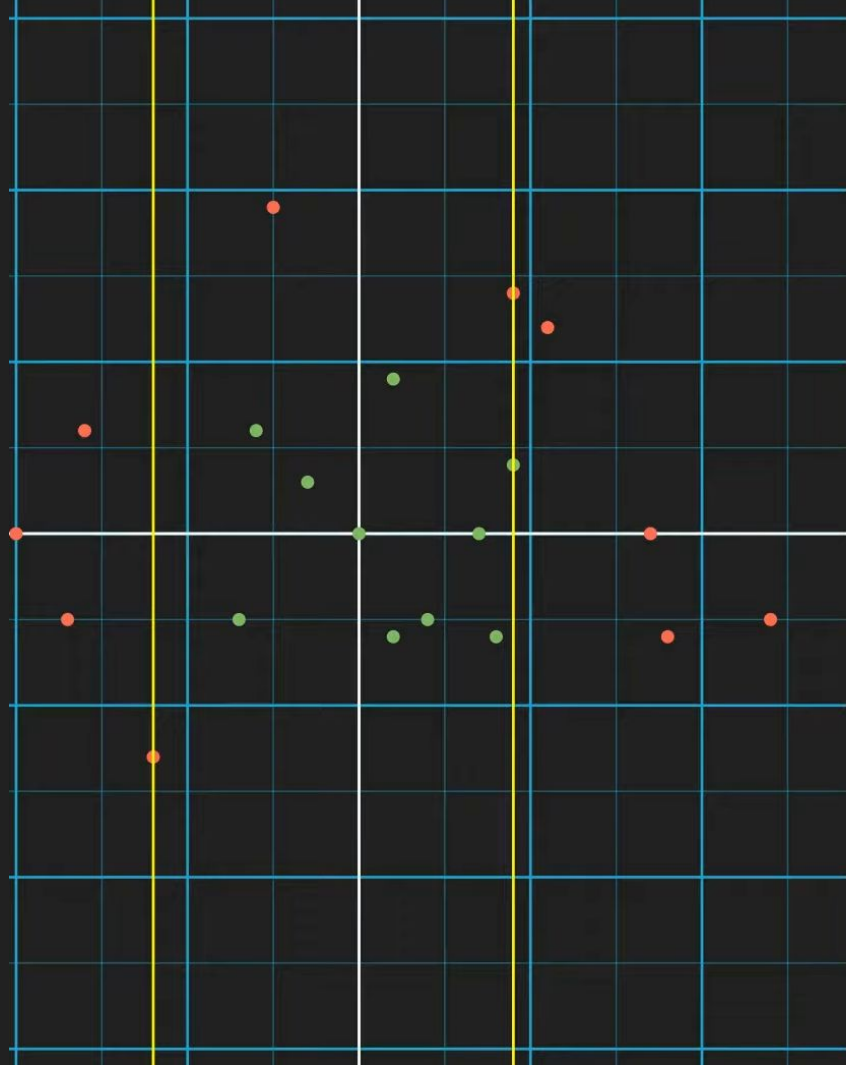


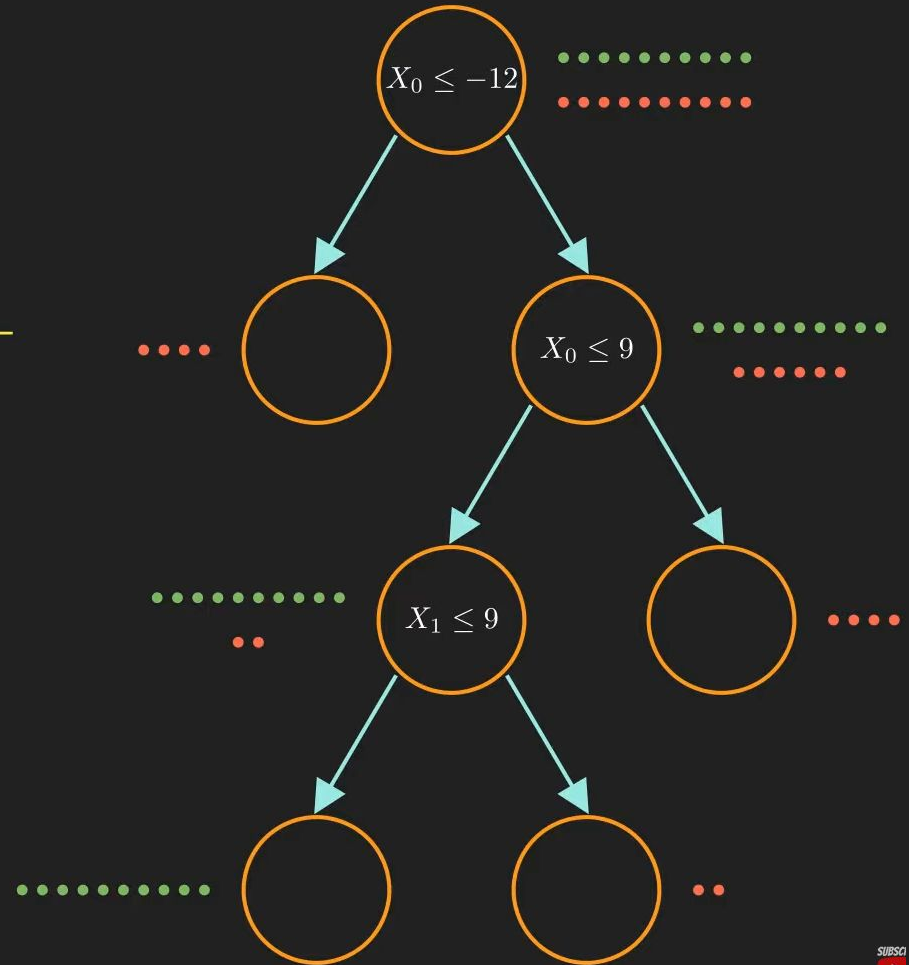
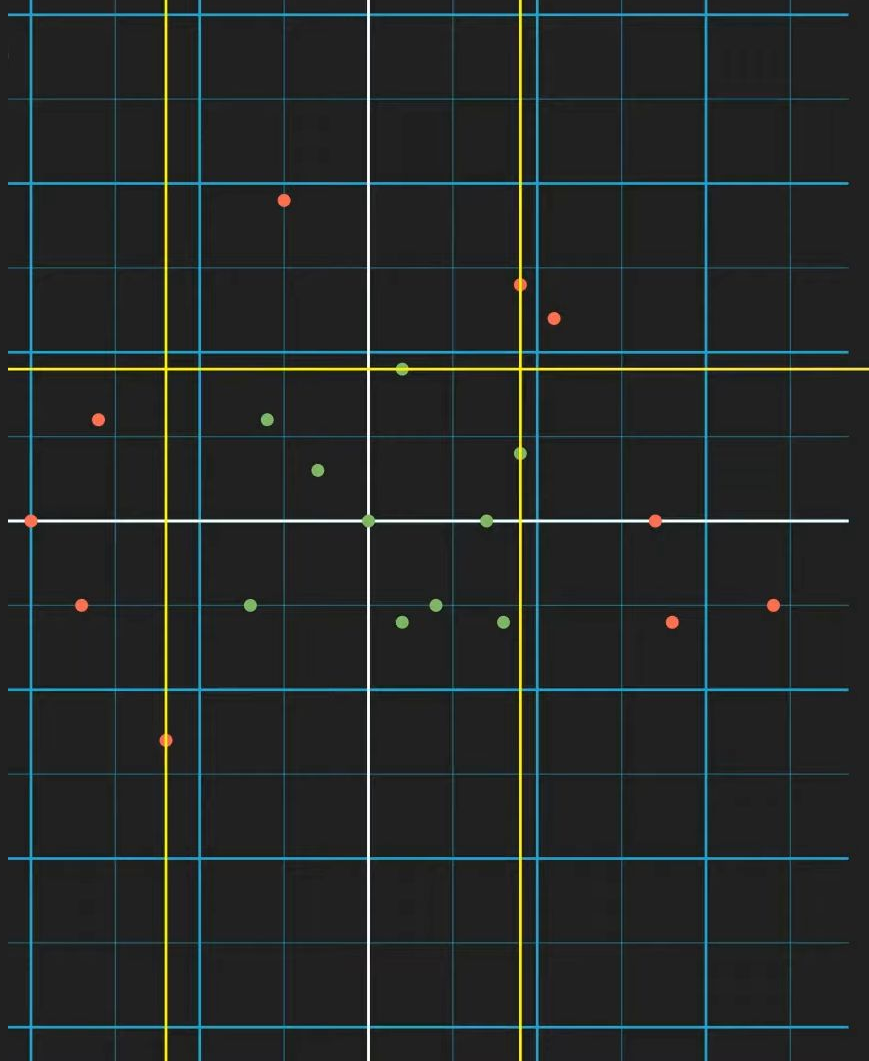
Not linearly separable :(

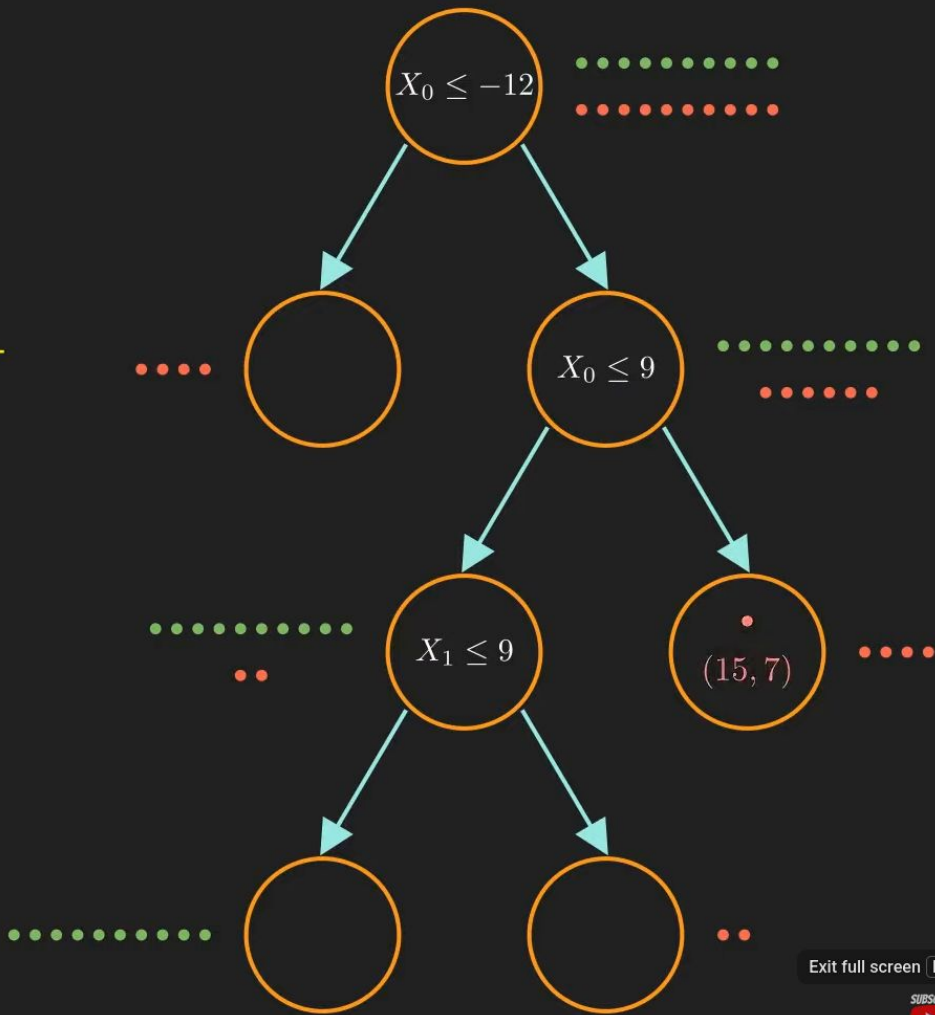
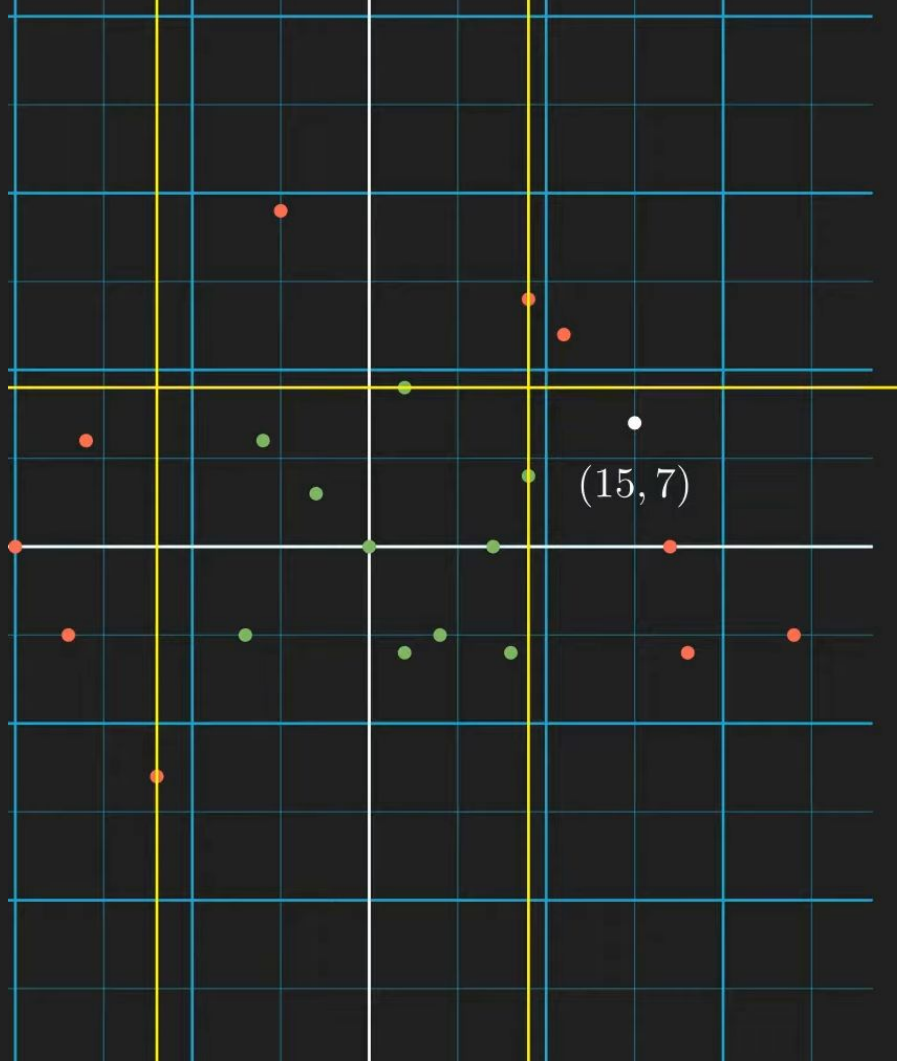


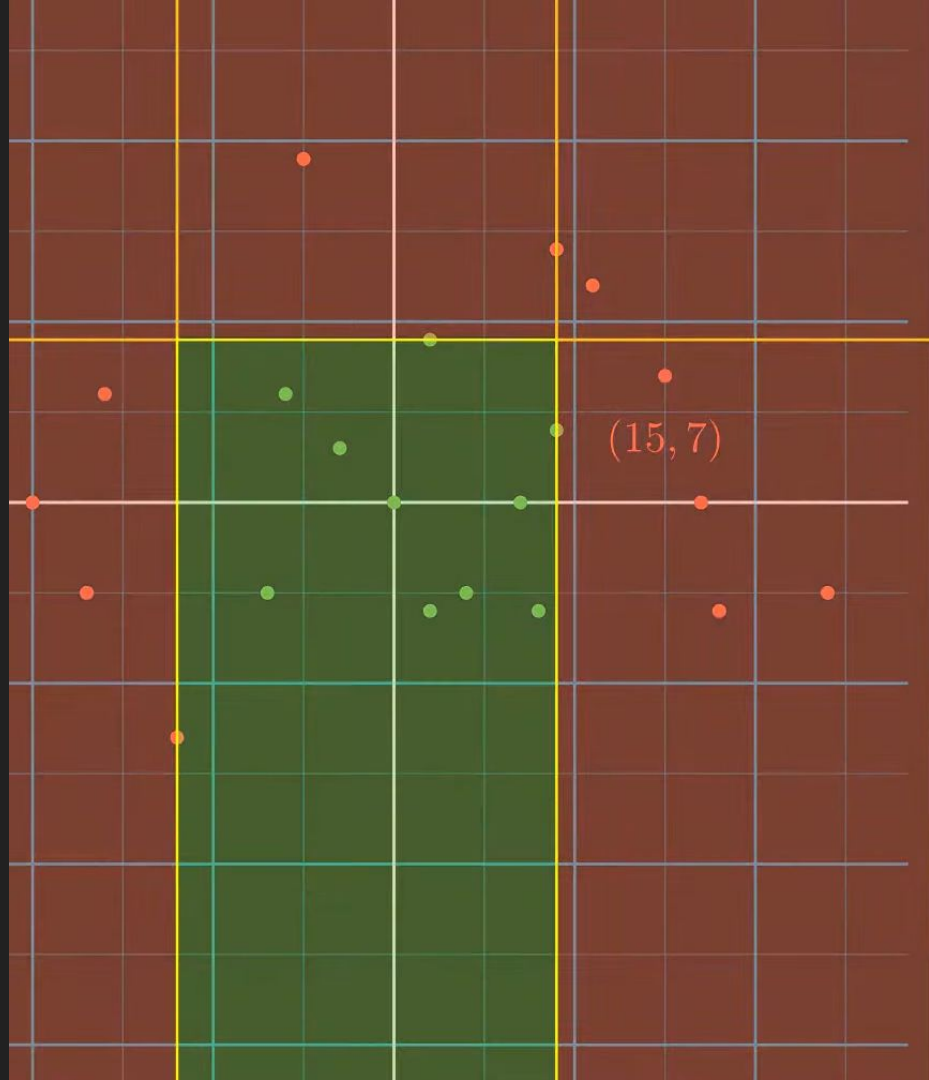


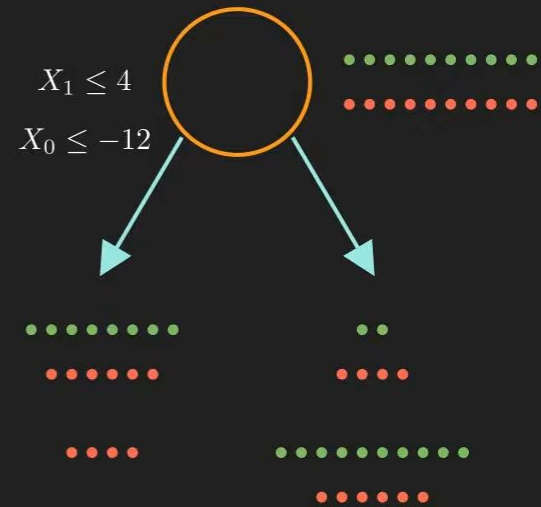
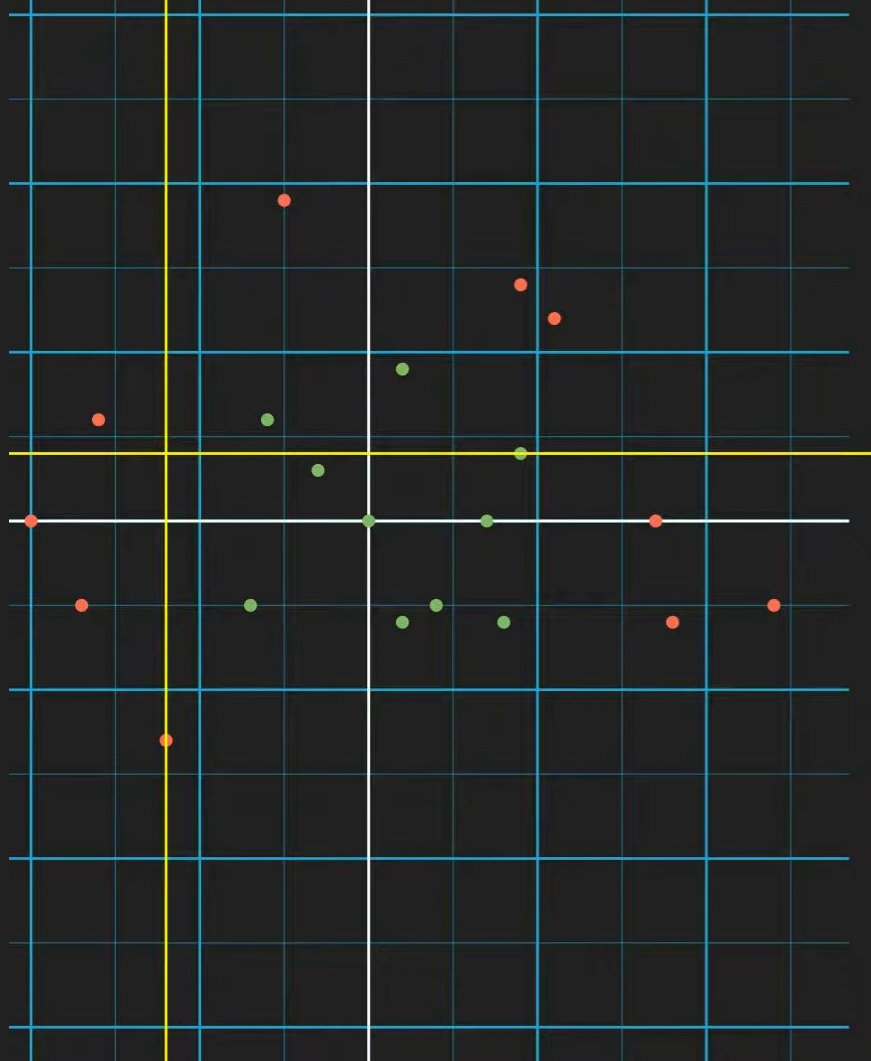






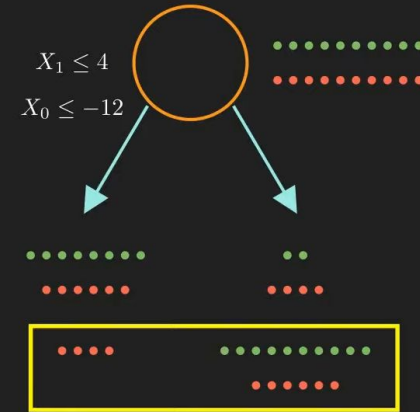
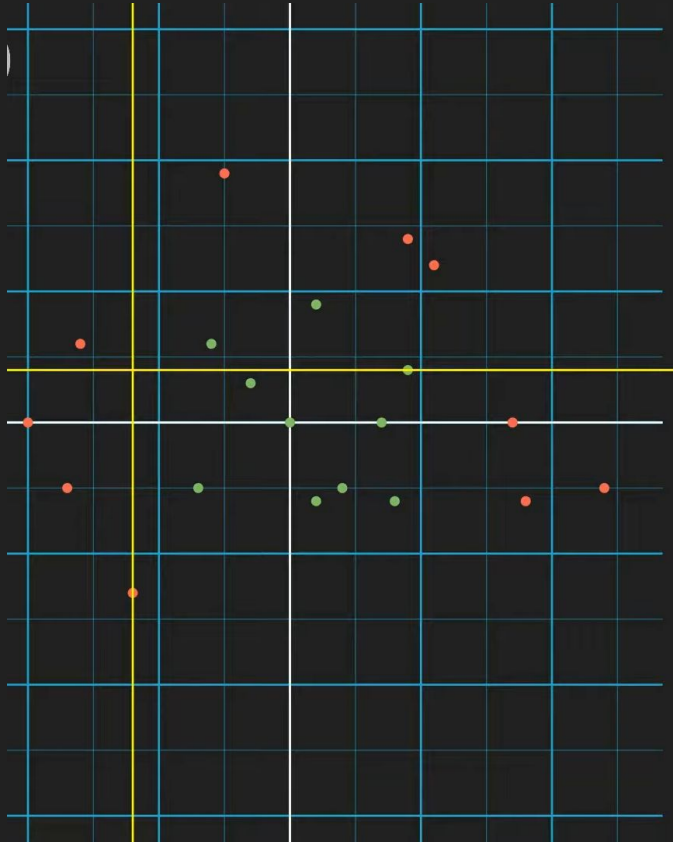






Which split is better?

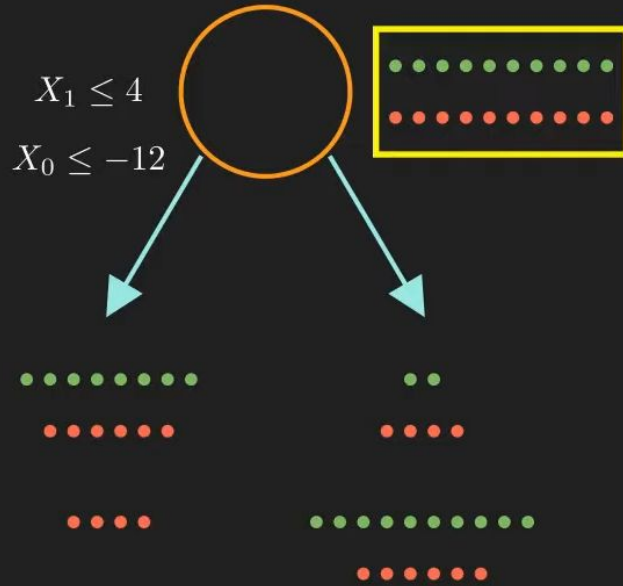
We need pure leaf nodes



Which split is better?

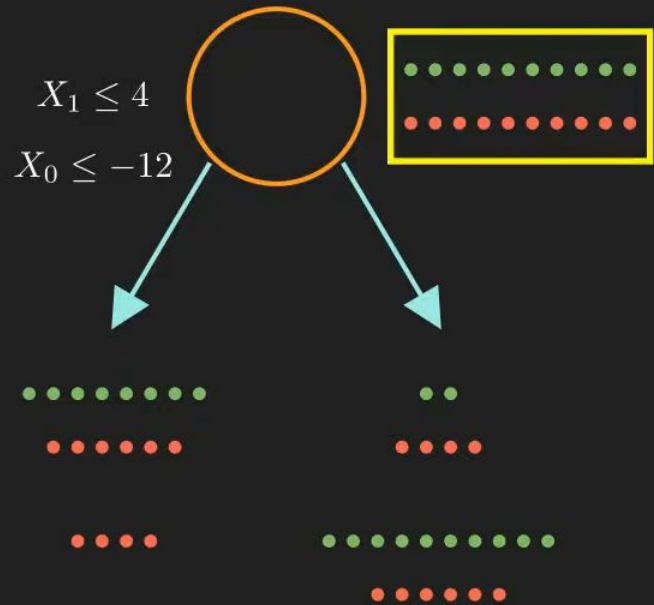
We will measure the impurity using the Entropy

$$H = - \sum_p p \log(p)$$

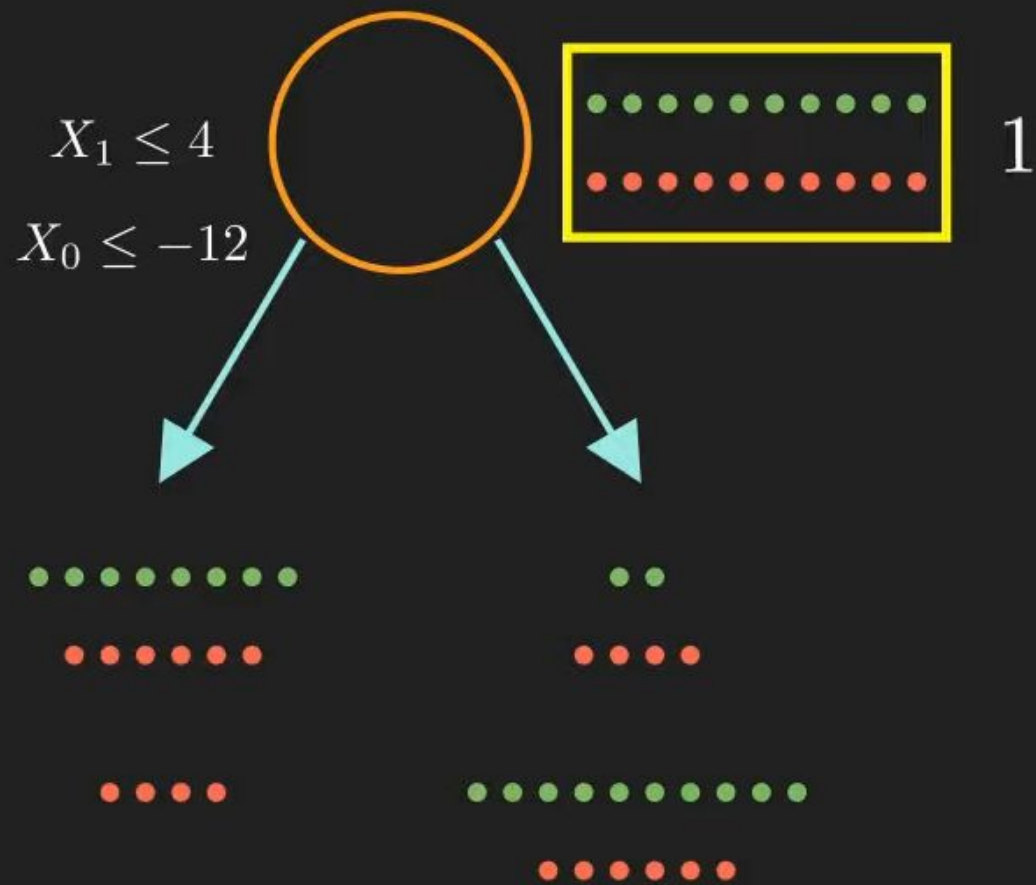


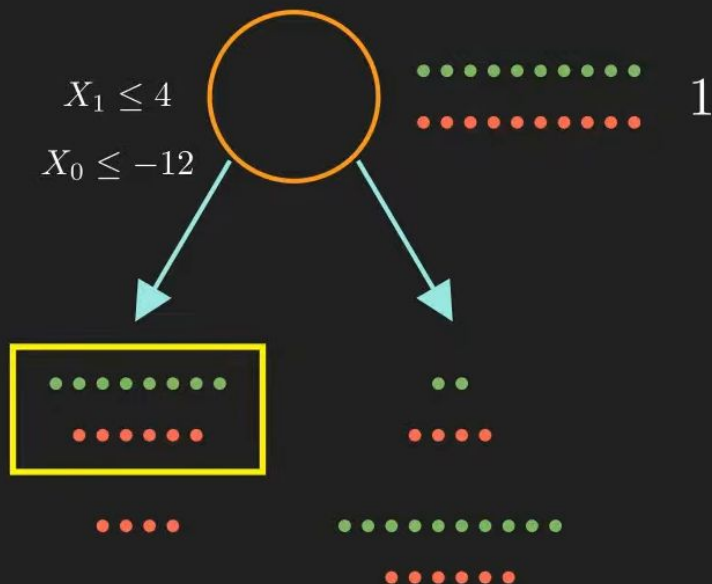
$$Entropy = \sum - p_i \log(p_i)$$

p_i = probability of class i

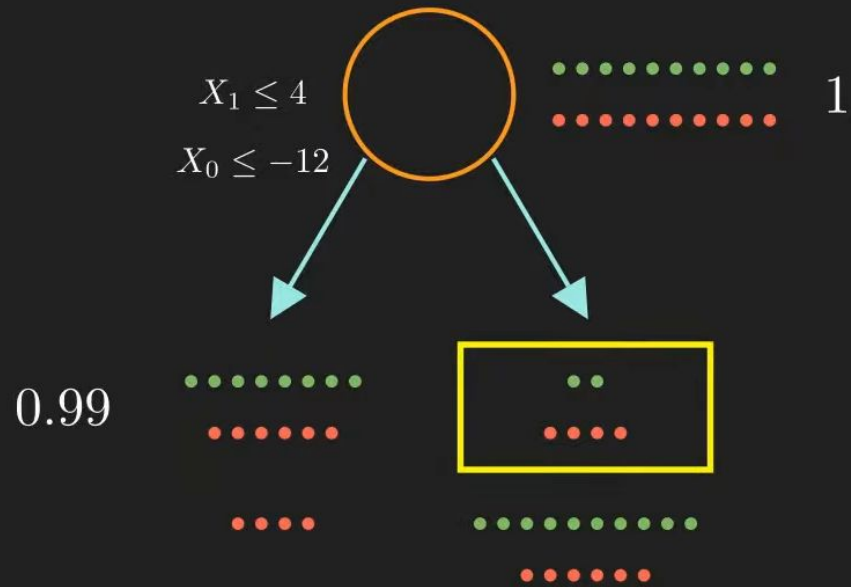


$$-0.5 \log(0.5) - 0.5 \log(0.5)$$

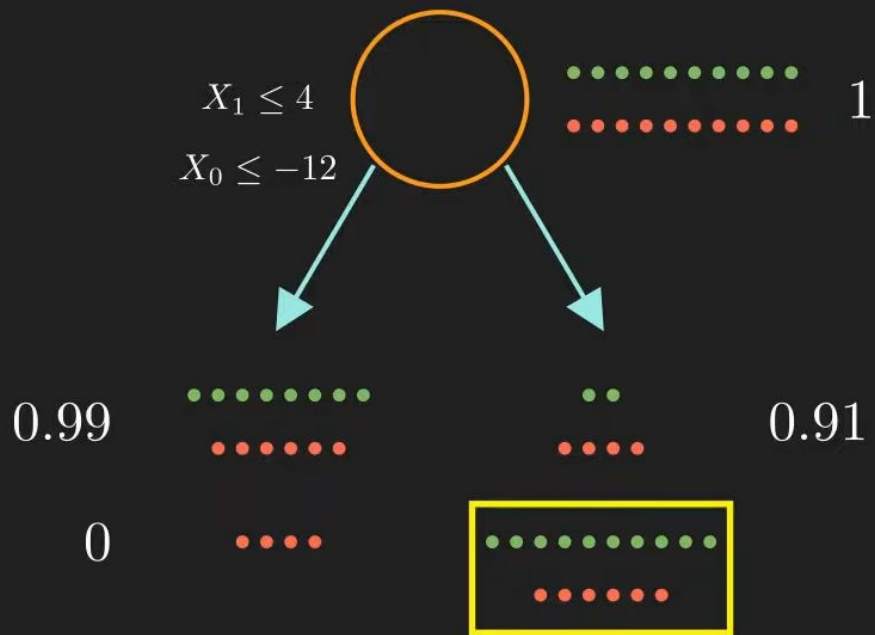




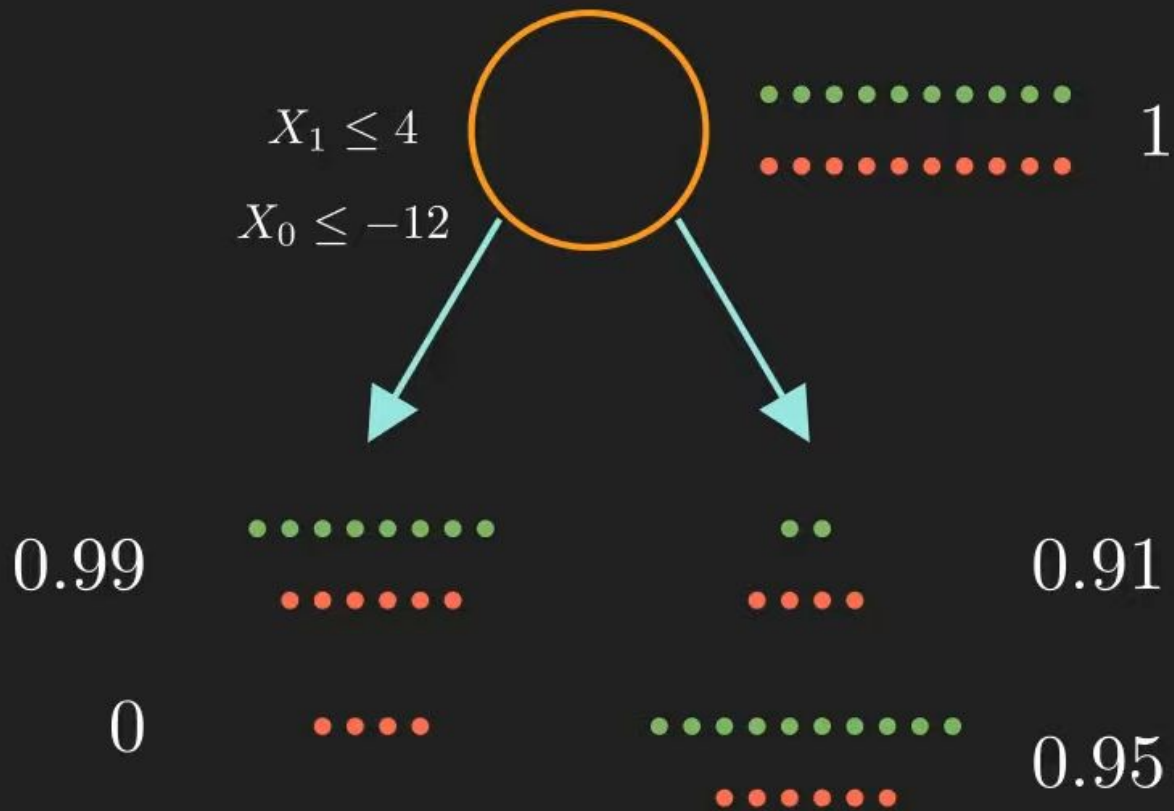
$$-0.57 \log(0.57) - 0.43 \log(0.43)$$

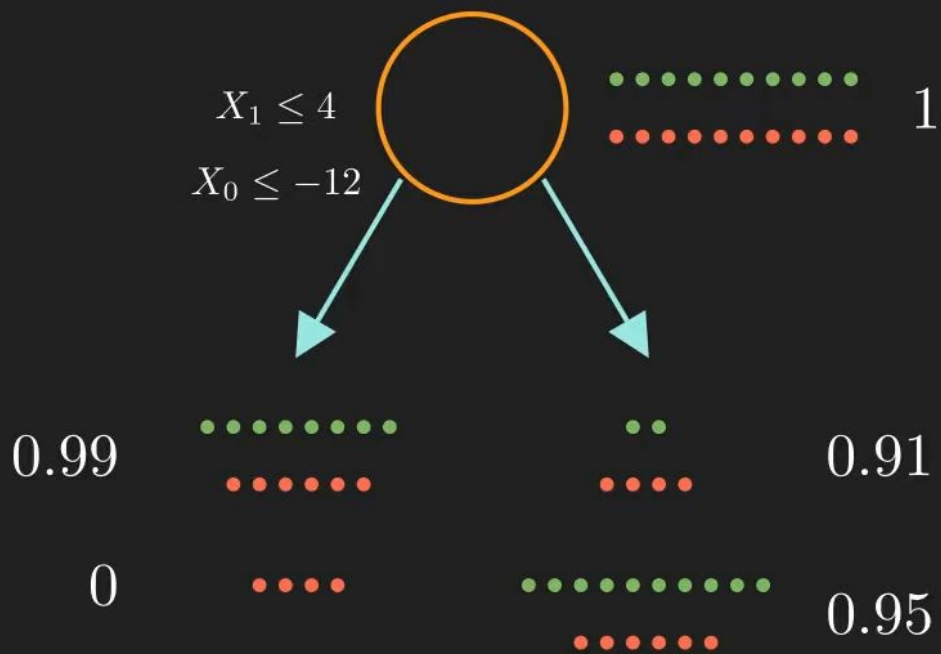


$$-0.33 \log(0.33) - 0.67 \log(0.67)$$



$$-0.63 \log(0.63) - 0.37 \log(0.37)$$



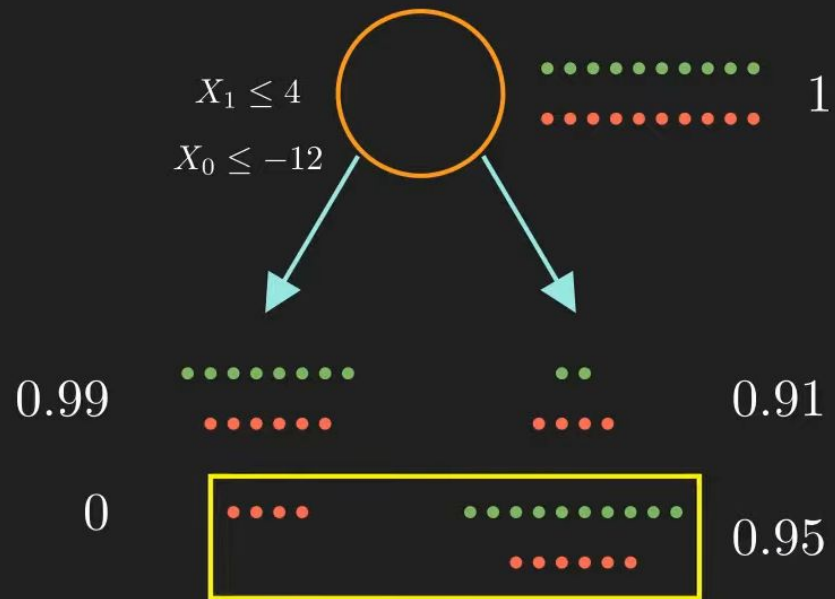


$$IG = E(parent) - \sum w_i E(child_i)$$

$$\text{IG} = E(\textit{parent}) - \sum w_i E(\textit{child}_i)$$

$$\text{IG}_1 = 1 - \frac{14}{20} \times .99 - \frac{6}{20} \times .91 = 0.034$$

$$\text{IG}_2 = 1 - \frac{4}{20} \times 0 - \frac{16}{20} \times .95 = 0.24$$



$$IG_2 > IG_1$$

Naive Bayes

features: $X = (X_1, X_2, \dots, X_n)$

label: Y

$$P(Y = y|X = (x_1, x_2, \dots, x_n))$$

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Likelihood

Prior

Posterior

The diagram illustrates Bayes' Theorem with the following components and arrows:

- Posterior** (green text) has a curved arrow pointing to the $P(Y|X)$ term in the equation.
- Likelihood** (red text) has a curved arrow pointing to the $P(X|Y)$ term in the numerator.
- Prior** (blue text) has a curved arrow pointing to the $P(Y)$ term in the numerator.
- Evidence** (yellow text) has a curved arrow pointing to the $P(X)$ term in the denominator.

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Evidence

Categorical Data

X_1	X_2	Y
0	0	0
0	1	1
1	2	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
2	1	0
1	0	0

$$X_1, X_2 \in \{0, 1, 2\}$$

$$Y \in \{0, 1\}$$

Estimate the value of Y given that $X = (0, 2)$

Let's compute $P(Y = 0|X = (0, 2))$ and $P(Y = 1|X = (0, 2))$...

X_1	X_2	Y
0	0	0
0	1	1
1	2	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
2	1	0
1	0	0

Let's compute $P(Y = 0|X = (0, 2))$ and $P(Y = 1|X = (0, 2))$...

$$P(Y = 0) = \frac{\#Y = 0}{\#Y = 0 + \#Y = 1} = \frac{6}{10}$$

$$P(Y = 1) = \frac{\#Y = 1}{\#Y = 0 + \#Y = 1} = \frac{4}{10}$$

X_1	X_2	Y
0	0	0
0	1	1
1	2	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
2	1	0
1	0	0

Let's compute $P(Y = 0|X = (0, 2))$ and $P(Y = 1|X = (0, 2))$...

$$P(Y = 0) = \frac{\#Y = 0}{\#Y = 0 + \#Y = 1} = \frac{6}{10}$$

$$P(Y = 1) = \frac{\#Y = 1}{\#Y = 0 + \#Y = 1} = \frac{4}{10}$$

$$P(X = (0, 2)|Y = 1) = \frac{1}{4}$$

$$P(X = (0, 2)|Y = 0) = 0$$

X_1	X_2	Y
0	0	0
0	1	1
1	2	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
2	1	0
1	0	0

Let's compute $P(Y = 0|X = (0, 2))$ and $P(Y = 1|X = (0, 2))$...

$$P(Y = 0) = \frac{\#Y = 0}{\#Y = 0 + \#Y = 1} = \frac{6}{10}$$

$$P(Y = 1) = \frac{\#Y = 1}{\#Y = 0 + \#Y = 1} = \frac{4}{10}$$

$$P(X = (0, 2)|Y = 1) = \frac{1}{4}$$

$$P(X = (0, 2)|Y = 0) = 0$$

X_1	X_2	Y
0	0	0
0	1	1
1	2	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
2	1	0
1	0	0

Note

Since $P(X_1, X_2, \dots, X_n)$ is constant across classes, we focus on maximizing the numerator

Let's compute $P(Y = 0|X = (0, 2))$ and $P(Y = 1|X = (0, 2))$...

$$P(Y = 0) = \frac{\#Y = 0}{\#Y = 0 + \#Y = 1} = \frac{6}{10}$$

$$P(Y = 1) = \frac{\#Y = 1}{\#Y = 0 + \#Y = 1} = \frac{4}{10}$$

$$P(X = (0, 2)|Y = 1) = \frac{1}{4}$$

$$P(X = (0, 2)|Y = 0) = 0$$

$$P(X = (0, 2)|Y = 0) * P(Y = 0) = 0 * \frac{6}{10} = 0$$

$$P(X = (0, 2)|Y = 1) * P(Y = 1) = \frac{1}{4} * \frac{4}{10} = \frac{1}{10}$$

X_1	X_2	Y
0	0	0
0	1	1
1	2	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
2	1	0
1	0	0

Estimated value of Y is 1 given $X = (0, 2)$

What if X not exists??!!!

Be Naive :)



Naive

Let's assume X_1, X_2 are independant!

Let's compute $P(Y = 0|X = (0, 2))$ and $P(Y = 1|X = (0, 2))$...

$$P(Y = 0) = \frac{\#Y = 0}{\#Y = 0 + \#Y = 1} = \frac{6}{10}$$

$$P(Y = 1) = \frac{\#Y = 1}{\#Y = 0 + \#Y = 1} = \frac{4}{10}$$

$$P(X = (0, 2)|Y = 1) = P(X_1 = 0|Y = 1) * P(X_2 = 2|Y = 1) = \frac{3}{4} * \frac{2}{4}$$

$$P(X = (0, 2)|Y = 0) = P(X_1 = 0|Y = 0) * P(X_2 = 2|Y = 0) = \frac{1}{6} * \frac{1}{6}$$

$$\frac{3}{4} * \frac{2}{4} > \frac{1}{6} * \frac{1}{6}$$

So, the estimated value of Y is 1

X_1	X_2	Y
0	0	0
0	1	1
1	2	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
2	1	0
1	0	0

Dealing with Continuous features

1. Discretization

$$age \in (11, 50) \longrightarrow age \in \{1, 2, 3, 4\}$$

2. Fit a known Distribution

GNB

$$P(x_i|y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Practical Example

To understand how Gaussian Naive Bayes works here's a simple binary classification problem using one feature: petal length.

Petal Length (cm)	Class Label
1.4	0 (Iris-setosa)
1.3	0 (Iris-setosa)
1.5	0 (Iris-setosa)
4.5	1 (Iris-versicolor)
4.7	1 (Iris-versicolor)
4.6	1 (Iris-versicolor)

We want to classify a new sample with **petal length = 1.6 cm**.

1. Separate by Class

- *Class0* : [1.4, 1.3, 1.5]
- *Class1* : [4.5, 4.7, 4.6]

2. Calculate Mean and Variance

For class 0:

- $\mu_0 = \frac{1.4+1.3+1.5}{3} = 1.4$
- $\sigma_0^2 = \frac{(1.4-1.4)^2+(1.3-1.4)^2+(1.5-1.4)^2}{3} = 0.0067$

For class 1:

- $\mu_1 = \frac{4.5+4.7+4.6}{3} = 4.6$
- $\sigma_1^2 = \frac{(4.5-4.6)^2+(4.7-4.6)^2+(4.6-4.6)^2}{3} = 0.0067$

3. Gaussian Likelihood

The Gaussian PDF is:

$$P(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

For $x = 1.6$:

Class 0

$$\bullet P(1.6|C = 0) \approx \frac{1}{\sqrt{2\pi \cdot 0.0067}} \cdot e^{-\frac{(1.6-1.4)^2}{2 \cdot 0.0067}} \approx 0.247$$

Class 1

$$\bullet P(1.6|C = 1) \approx \frac{1}{\sqrt{2\pi \cdot 0.0067}} \cdot e^{-\frac{(1.6-4.6)^2}{2 \cdot 0.0067}} \approx 0$$

4. Multiply by Class Priors

Assume equal priors:

$$P(C = 0) = P(C = 1) = 0.5$$

Then:

- $P(C = 0|x) \propto 0.247 \cdot 0.5 = 0.1235$
- $P(C = 1|x) \propto 0 \cdot 0.5 = 0$

5. Prediction

Since $P(C = 0|x) > P(C = 1|x)$,

Predicted Class: $C = 0$ (Iris-setosa)

